ABSOLUTE FROBENIUS ACTS TRIVIALLY ON COHOMOLOGY WITH CONSTANT TORSION COEFFICIENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file trace.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Lemma lemma-baffling: a self-map g of X for which every etale U -> X admits a functorial isomorphism U \cong U \times_{varphi,X,g} X over X induces the identity on cohomology (author lemma environment, source lines 299-315). the theorem is proved by verifying this lemma's hypothesis, as the author states in the carried bridge sentence at source lines 340-343. Reduces the theorem to the functorial-isomorphism construction carried out in the proof.
- Etale Morphisms, Lemma etale-lemma-relative-frobenius-etale (canonical tag 0EBS): the relative frobenius of an etale morphism is etale and a universal homeomorphism. cited at proof line 359 for the dotted arrow in the diagram. Supplies the key fact that the dotted arrow is an isomorphism, completing the verification of the lemma's hypothesis.

Source reference. Original lemma 03SP: sources/stacks-chow/tags.txt, source lines 4976–4976. The theorem is deduced by verifying this lemma's hypothesis (announced at lines 340-343, used at proof lines 344-359); the carried bridge sentence cites it.

Source reference. Original lemma 0EBS: sources/stacks-chow/tags.txt, source lines 17932–17932. Cited at proof line 359 to conclude the dotted arrow is an isomorphism; load-bearing inside the counted span.

Theorem 0.2 (The Baffling Theorem)**.** *Let X be a scheme in characteristic $p > 0$. Then the absolute frobenius induces (by pullback) the trivial map on cohomology, i.e., for all integers $j \geq 0$,*

$$F_X^* : H^j(X, \underline{\mathbf{Z}/n\mathbf{Z}}) \longrightarrow H^j(X, \underline{\mathbf{Z}/n\mathbf{Z}})$$

is the identity.

To prove the theorem, we merely verify that the assumption of Lemma 03SP holds for the frobenius.

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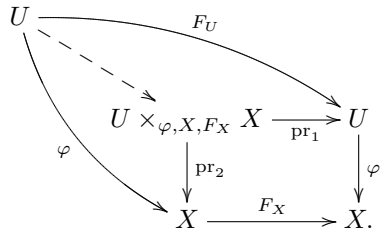
Proof of Theorem 0.2. We need to verify the existence of a functorial isomorphism as above. For an étale morphism $\varphi : U \to X$, consider the diagram



The dotted arrow is an étale morphism and a universal homeomorphism, so it is an isomorphism. See Étale Morphisms, Lemma 0EBS. □